

AUC in Plain Language

A visual companion to Module 3, Slide 8

Case: pediatric chest X-ray prioritization for human review

One-sentence meaning: AUC summarizes how well a model puts pneumonia-labeled images above normal-labeled images when it ranks them by score.

The answer first

The Module 3 CNN has a test **ROC AUC of 0.960** on 624 untouched test images.

In plain language:

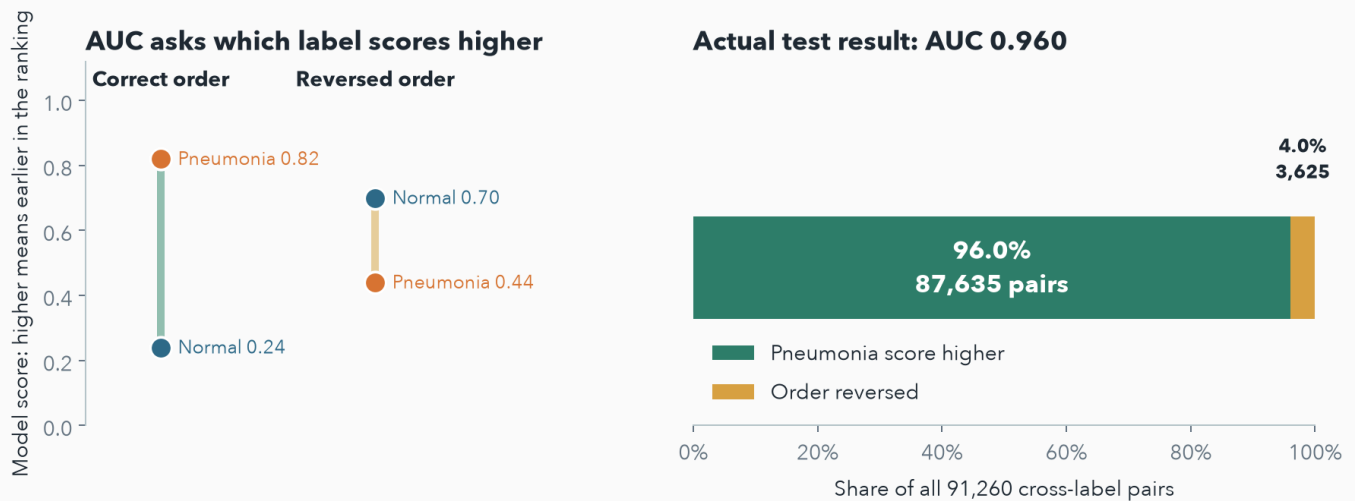
- Choose one pneumonia-labeled image and one normal-labeled image from the test set.
- Compare their model scores.
- The pneumonia-labeled image receives the higher score about **96 times out of 100**.

Across every possible cross-label pair in this test set:

Item	Measured count
Pneumonia-labeled images	390
Normal-labeled images	234
Pairs compared: 390 x 234	91,260
Pneumonia-labeled image scored higher	87,635
Normal-labeled image scored higher	3,625
Tied scores	0

$$\text{AUC} = (87,635 + 0.5 \times 0) / 91,260 = 0.960278$$

AUC is a ranking summary, not a diagnosis



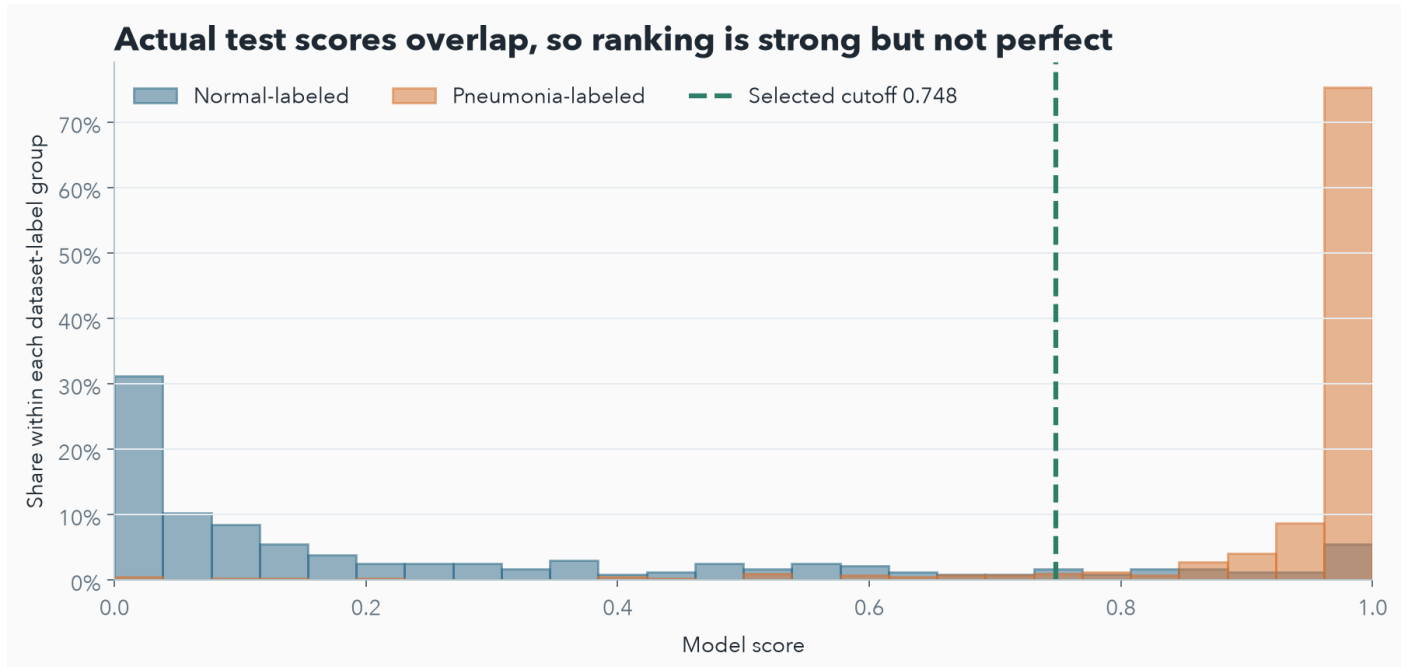
Read the chart: A correct pair does not mean both images were diagnosed. It means only that the image carrying the pneumonia dataset label received the higher ranking score.

Do not say: "The model is 96% accurate" or "this child has a 96% probability of pneumonia." Neither statement follows from AUC.

1. Start with scores, not decisions

The CNN gives every image a score between 0 and 1. In this notebook, a higher score means the image looks more like pneumonia-labeled training examples **to this model**.

The scores are used for ranking. They were not calibrated as patient-level probabilities.



How to read this chart

- The horizontal axis is the model score.
- Each colored shape shows the **share within that label group** at each score range.
- Orange is the pneumonia-labeled group; blue is the normal-labeled group.
- The groups overlap. In that overlap, some normal-labeled images score above some pneumonia-labeled images.
- Less overlap usually means better ranking and a larger AUC.
- The green dashed line is the selected operating cutoff, not part of the AUC calculation.

Why the overlap matters

If every orange score were above every blue score, AUC would be **1.0**. If the two groups were mixed with no useful ordering, AUC would be around **0.5**. This model falls between those endpoints and is much closer to perfect ranking on this particular test set.

The chart also explains why AUC is not perfection. The overlapping scores create reversed pairs and lead to false positives or false negatives once a cutoff is selected.

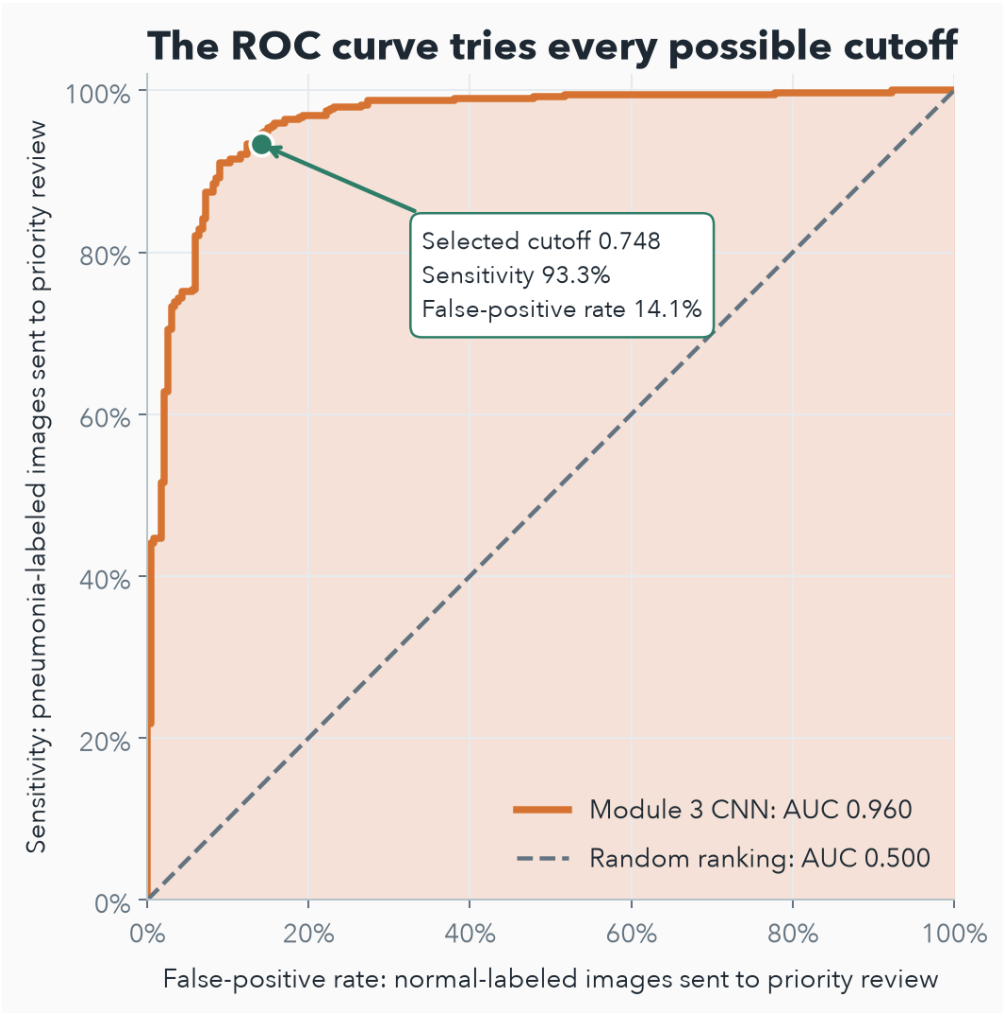
2. What is an ROC curve?

ROC stands for **receiver operating characteristic**. The historical name matters less than the question the curve answers:

What happens to correct prioritization and false alarms as we try every possible cutoff?

For each candidate cutoff, calculate two percentages:

Axis	Plain-language question	Formula
Vertical: sensitivity or true-positive rate	Of all pneumonia-labeled images, what share enters priority review?	$TP / (TP + FN)$
Horizontal: false-positive rate	Of all normal-labeled images, what share also enters priority review?	$FP / (FP + TN) = 1 - \text{specificity}$



How to read this chart

- Every point on the orange curve represents a different cutoff.
- The bottom-left corner means almost nothing is prioritized.
- The top-right corner means almost everything is prioritized.
- A useful curve bends toward the top-left: high sensitivity with fewer false positives.
- The diagonal line represents random ranking on average; its AUC is **0.5**.
- The shaded area under the orange curve is the AUC.

At the selected cutoff of **0.7480410**, the test-set operating point is:

- **Sensitivity:** $364 / 390 = 93.3\%$
- **False-positive rate:** $33 / 234 = 14.1\%$
- **Specificity:** $201 / 234 = 85.9\%$
- **Priority-review share:** $397 / 624 = 63.6\%$

Those four numbers describe one selected cutoff. The AUC summarizes ranking across **all** cutoffs.

Remember: the point describes one policy cutoff; the shaded area summarizes ranking across all possible cutoffs.

3. AUC and the confusion matrix answer different questions

Measure	Needs one cutoff?	Main question
ROC AUC	No	How well do scores rank one label above the other across possible cutoffs?
Sensitivity	Yes	Of pneumonia-labeled images, how many enter priority review?
Specificity	Yes	Of normal-labeled images, how many remain in standard review?
Precision	Yes	Of priority-review images, how many carry the pneumonia label?
Priority-review share	Yes	How much of the complete queue is sent to priority review?
Confusion matrix	Yes	How many images land in each correct or incorrect route?

Why AUC cannot choose the cutoff for us

AUC tells us whether the score is useful for ranking. It does not tell us how many false negatives are acceptable, how much priority capacity exists, or what delay each route creates.

The cutoff is therefore a **policy choice informed by evidence**, not something contained inside the AUC. Module 3 selects its cutoff on validation data to meet a sensitivity objective while avoiding a priority queue that contains every image.

AUC is not accuracy

Accuracy counts correct decisions after selecting a cutoff:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{FP} + \text{FN} + \text{TN})$$

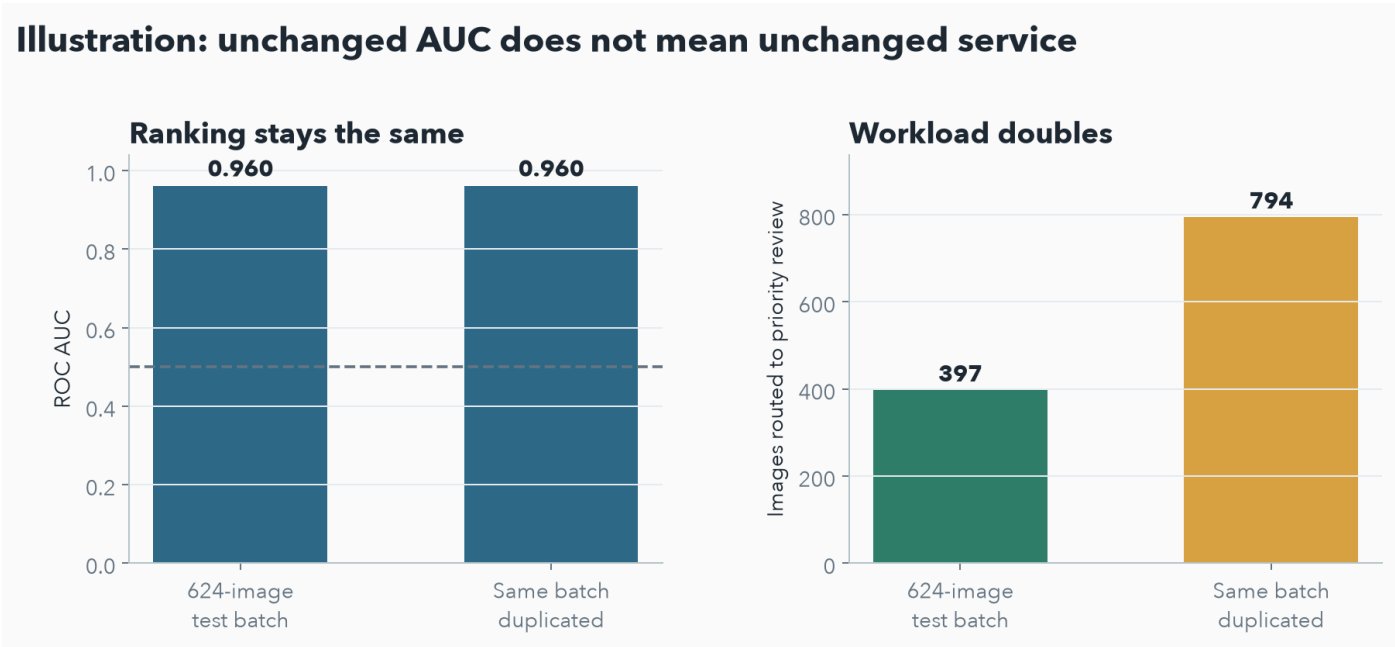
The Module 3 test accuracy is **90.5%** at the selected cutoff. Its AUC is **0.960**. They are different numbers because they answer different questions.

AUC is not calibration

A model can rank images correctly while its numeric scores are too high or too low. AUC may remain unchanged if every score is transformed in a way that preserves the order. Therefore, **0.80** should not be described as an 80% probability unless probability calibration has been evaluated separately.

4. AUC can stay steady while the service gets worse

Imagine the same mix of images and the same model-score ordering, but twice as many studies arrive. Duplicating the batch preserves the AUC because the ranking pattern is unchanged. The number of studies entering priority review doubles.



What this chart proves

- Baseline test batch: 624 images, 397 routed to priority review.
- Duplicated batch: 1,248 images, 794 routed to priority review.
- AUC remains **0.960** because the scores and labels have the same ordering pattern.
- Workload doubles because the service must handle twice as many studies.

This is why Slide 8 separates three monitoring layers:

Layer	Examples of what to monitor
Model	AUC, sensitivity, calibration, drift, image quality
Workflow	Queue volume, time to read, overrides, alerts acted on
Patient	Time to interpretation, access, outcomes, subgroup differences

AUC can look stable while the priority queue grows, staff cannot act quickly enough, or one subgroup performs worse than the aggregate.

5. What AUC tells us, and what it does not

AUC does tell us

- How well scores separate two dataset labels in ranking order.
- Performance across all candidate cutoffs rather than one selected cutoff.
- Whether ranking is better than random on the evaluated data.
- A useful summary for comparing ranking models on the same task and evaluation design.

AUC does not tell us

- Whether this child has pneumonia.
- Whether a score is a calibrated probability.
- Which cutoff the healthcare workflow should use.
- How many images enter the priority queue at the selected cutoff.
- How long patients wait for interpretation.
- Whether performance is safe at another hospital, on another device, or in another population.
- Whether performance is similar across important subgroups.
- Whether the model improves patient outcomes.

6. A practical reading checklist

When someone reports an AUC, ask:

1. **What was ranked?** Name the positive and negative labels.
2. **Which data produced it?** Training, validation, or untouched test data?
3. **How many examples and sites?** A single aggregate may hide uncertainty or variation.
4. **Was the test population appropriate?** Age, setting, device, prevalence, and protocol matter.
5. **Which cutoff will be used?** Then inspect the confusion matrix and workload at that cutoff.
6. **Can the workflow act?** Monitor queue volume and time to interpretation.
7. **What remains human-owned?** Here, radiologists interpret images and clinicians diagnose and treat.

7. Mini glossary

Term	Plain-language meaning
Score	A number the model uses to rank images.
Cutoff / threshold	The score boundary that converts ranking into a route.
ROC curve	All sensitivity and false-positive-rate tradeoffs created by trying different cutoffs.
AUC	The area under that ROC curve; equivalently, cross-label pair-ranking performance.
Sensitivity	Share of pneumonia-labeled images routed to priority review.
Specificity	Share of normal-labeled images kept in standard review.
False-positive rate	Share of normal-labeled images routed to priority review; $1 - \text{specificity}$.
Calibration	Whether numeric scores match observed frequencies well enough to be read as probabilities.

Data and sources

Measured course artifacts: PneumoniaMNIST 128 test split, 624 images; frozen model and policy from the Module 3 notebook. Test AUC **0.9602783257** ; cutoff **0.7480410337** ; TN/FP/FN/TP **201/33/26/364** .

References:

1. scikit-learn, [ROC AUC score documentation](#).
2. scikit-learn, [Receiver operating characteristic metrics](#).
3. Fawcett, T. (2006), [An introduction to ROC analysis](#), *Pattern Recognition Letters*, 27(8), 861-874.

Educational boundary: This guide explains evaluation of a packaged classroom model. It does not establish clinical validity and must not be used for patient diagnosis or treatment.